

Canonical Flexible Transition-state Theory for Generalized Reaction Paths

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In previous work (*J. Chem. Phys.*, 1995, **103**, 2917), simple yet exact formulae for the canonical flexible transition-state theory expression for the thermal reaction-rate constant were derived for all pairings of atomic, linear rigid top, and non-linear rigid top fragments when the distance between the centres of mass of the fragments serves as the reaction coordinate. In this paper, we derive the fundamental modifications required to generalize the reaction coordinate for all fragment-type pairings. That is, the hinge point about which each fragment rotates is no longer constrained to be the centre of mass of the fragment, being in general arbitrarily displaced from the fragment centre of mass. The generalized reaction coordinate is the line connecting the displaced hinge points. It is shown that only the kinetic energy associated with the internal relative motion of the two fragments is affected by a generalized reaction coordinate. The correction to this kinetic energy has a simple functional form whose evaluation for a given fragment-type pairing is straightforward. The ensuing formulae for the canonical rate constant will not be as simple as for the centres-of-mass reaction coordinate case, but are not more difficult to employ computationally. The new theory is applied to the simplest possible fragment-type pairing, atom–diatom, which serves to illustrate the essential features of the method and some of the implications of a generalized reaction coordinate.

1. Introduction

Reactions, either dissociative or associative, whose transition states or ‘bottlenecks’ are characterized by large amplitude motion, constitute a large and important class of chemical processes. This situation most commonly arises when there is no pronounced potential barrier for the formation of a parent molecular from a pair of constituent molecular fragments. Such ‘barrierless’ reactions are commonplace in chemistry, particularly in combustion, atmospheric, and interstellar processes. Large amplitude transition-state motion can also occur in cases where there is a barrier, for example, in

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heavy–light–heavy reactions such as $\text{Cl}^\bullet + \text{HCl}$, where the bending degree of freedom at the transition state might be thought of as a hindered rotation of HCl (*e.g.*, ‘figure eight’ reaction trajectories).¹ Despite ongoing advances in computer technology, classical trajectories and quantum dynamics, the most widespread theories for modelling reaction-rate coefficients are statistical theories. Such theories are generally one of two interrelated but distinct types: variational transition-state theory (VTST) and the quantum adiabatic channel model (QACM). The incorporation of large amplitude motion in QACM has been the subject of a recent study.² Here we present further developments for accurately incorporating such large amplitude motion in VTST.

The first version of VTST to account satisfactorily for large amplitude motion was flexible transition-state theory (FTST), which was introduced by Wardlaw and Marcus *circa* 1984.³ Like all variational theories, FTST allows the transition state to be mobile and not associated with any particular potential-energy surface (PES) feature; it typically occurs at larger interfragment separations for low energies and angular momenta and at smaller separations for high energies and angular momenta. However, FTST also treats the transition state as floppy and not associated with any particular geometry; there are large amplitude motions (transitional modes) which are coupled to each other and to overall rotation of the molecular system. Typically, transitional modes correspond to free rotations of the separated fragments that evolve into vibrational motions of the parent molecule.

The strength of the original FTST is the avoidance of approximations concerning the transitional modes by inclusion of the full transitional mode potential (or, the entire PES, if available) and by an exact enumeration of the transitional mode contribution to the transition-state sum of states. This enumeration can be feasibly accomplished only by a classical treatment of the transitional modes through phase-space integrals, which neglects quantum effects (such as zero-point energy) and which makes an individual variational location of the reaction bottleneck for each quantum state impossible. In contrast, QACM treats the transitional-mode potential in a highly approximate manner to achieve a quantum mechanical accounting of all levels in the transition state. A thorough discussion of the advantages, limitations, and approximations of original FTST is provided in recent review articles.⁴ Subsequent fundamental developments in FTST have been made by Aubanel and Wardlaw,⁵ Klippenstein and Marcus,⁶ Klippenstein,^{7–11} Smith,^{12–15} Aubanel *et al.*,¹⁶ and the present authors.¹⁷ This paper will blend two different improvements to FTST put forward by ourselves¹⁷ and Klippenstein.⁸

The price to be paid for inclusion of the potential-energy surface in FTST is complexity of implementation, intensive numerical calculations and loss of physical insight into the effect of PES features on the rate coefficient. These impediments prevent its routine use, *e.g.* in complex kinetic schemes consisting of many elementary steps involving radicals, and its casual use to interpret trends in the dependence of rate coefficients on PES features, temperature, size and type of radicals, *etc.* It is therefore desirable to develop versions of the original FTST which are easier to implement and are physically more transparent, yet which retain the essential features. Significant improvements along these lines have been made by Klippenstein⁹ and by Smith¹³ at the microcanonical level, and by ourselves¹⁷ at the canonical level. An explicit connection between Smith’s approach and our approach is established in Ref. 17. A common feature of all three approaches is that the momentum integrals in the classical phase-space expression for the sum of states along the reaction path can be done analytically. The remaining coordinate space integrals must, in general, be done numerically. The advantage of Smith’s¹³ and our¹⁷ reformulations of FTST is twofold: (1) The dimensionality of the integral expression for the sum of states is reduced by a factor of two compared with the corresponding original FTST expression; (2) the variables in the coordinate space integral are Jacobi internal angles which describe the transitional modes by specifying the orientation of fragments

relative to the vector R_{cm} connecting the centre of mass of the two fragments. We point out that use of R_{cm} as the reaction coordinate is implicit in the treatments of Ref. 13 and 17. The clear relationship between these angles and bond internal coordinates considerably reduces the effort devoted to the transformation of coordinates for the purpose of evaluating the PES which is most often expressed in bond coordinates. Indeed, the PES may be expressed in terms of Jacobi coordinates in the first place, thus eliminating the need for a coordinate transformation. In original FTST, on the other hand, the sum of states is expressed as an integral over angular momenta and their conjugate angles; a complicated transformation is required to go from these variables to bond internal coordinates.

In a recent paper¹⁷ (hereafter denoted part I) we derived a new version of canonical FTST (CFTST) which is easier to implement, involves less computation, and provides more physical insight than the original FTST. Application of this theory provides an expression for the thermal rate coefficient, $k(T)$. The canonical approach requires only one flux minimization along the reaction coordinate for each temperature, as opposed to the (correct) microcanonical approach in which $N_{E,J}$ is first minimized for each energy and angular momentum, and then summed over J and Boltzmann-averaged over E to obtain a transition-state partition function at each temperature. A canonical estimation of the rate constant must always be higher than the true microcanonical one. For radical–radical association reactions for which this error has been studied, the overestimation of $k(T)$ by a canonical treatment is typically 5–20%, and is weakly temperature dependent.⁸ For ion–molecule reactions, the overestimation is expected to be larger, and for a particular case, is known to exceed 50% at some temperatures.⁹ Although our treatment in part I is formally equivalent to a canonical extension of Smith's microcanonical treatment, it is distinctive in that no reference is made to a microcanonical formulation and some of the physically important features of flexible transition states are presented in a conceptually appealing manner.

As it stands, our new version of CFTST is restricted to the reaction coordinate being the centres-of-mass separation R_{cm} . Extensive work by Klippenstein^{7,8} within the microcanonical FTST framework has revealed the limitations of using R_{cm} as the reaction coordinate for a handful of representative systems. Klippenstein has shown that use of a traditional bond reaction coordinate (*i.e.* atom–atom separation distance) can dramatically lower the corresponding sum of states at intermediate and small interfragment separations. Moreover, he has introduced the concept of an optimized reaction coordinate and implemented this idea for a restricted set of reaction coordinate variations. The motivation for the present paper is to incorporate a completely general reaction coordinate at the canonical level of FTST, the precise choice of reaction coordinate being optimized, at a given temperature and for a given system, by the usual variational minimization of the sum of states orthogonal to this coordinate. We do so by extending our new version of FTST, thus retaining the essential features and advantages of that treatment. In particular, the present treatment of generalized reaction paths within the framework of CFTST, and indeed within the broader context of canonical VTST, makes no reference to a microcanonical formulation and appears to be the first at the canonical level. Furthermore, the extension is formulated so that the resulting formulae and their numerical implementations involve relatively minor modifications of the existing centres-of-mass reaction coordinate treatment described in part I. The conceptual and computational advantages of the present treatment become especially important when employing non-traditional reaction coordinates whose optimization requires computational effort absent in a sum of states calculation using a traditional (non-optimized) reaction coordinate, such as a bond coordinate or the centre-of-mass coordinate.

The rest of this paper is divided as follows. Our canonical reformulation of part I is reviewed in Section 2. The extension to arbitrary reaction coordinates is described in Section 3, which is the focal point of the paper. Case-specific formulae for a simple

system (atom–diatom) are derived in Section 4. The complete range of case-specific formulae and selected numerical examples are deferred to a subsequent publication. The results of Section 4 are discussed in Section 5 and the paper concludes with a summary in Section 6.

2. Review of CFTST for the Centre-of-mass Reaction Path

2.1 Degrees of Freedom

Before giving explicit results from part I, it is useful to review the allocation of degrees of freedom for the various combinations of fragment types. A summary is provided in Table 1. All fragments are modelled as one of the following types of rigid tops: atomic, linear or non-linear. The total number of atoms in both fragments is N . The number of internal angles, $N_{q_{\text{int}}}$, describing the relative orientation of the two (rigid) fragments, is equal to the number of transitional modes. One coordinate, the reaction coordinate, specifies the separation of the fragments. There are three external rotation coordinates and $N_q = N_{q_{\text{int}}} + 3$ transitional/external rotational coordinates ($q = q_1, \dots, q_{N_q}$) in terms of which one constructs a kinetic-energy operator in matrix form. In our canonical formulation of the thermal rate constant in CFTST, the determinant of this matrix carries all relevant information about the kinetic energy for the transitional/external rotational modes. The number of conserved modes, $N_{\text{cons}} = 3N - 3 - (N_q + 1)$, is equal to the number of vibrational degrees of freedom in the two fragments. Table 1 also reveals that the problem under consideration is a finite one. That is, there are only five different cases to consider and, once case-specific formulae for a general reaction coordinate are obtained for each, the overall problem is solved.

2.2. Summary of Results

The canonical VTST expression for the rate coefficient is (with $\beta = 1/kT$)

$$k(T) = g_e \frac{1}{\beta h} \frac{\sigma}{\sigma^\ddagger} \frac{Q_{\text{TS}}^\ddagger}{Q_{\text{react}}(T)} \exp(-\beta V^\ddagger) \quad (1)$$

The quantity $Q_{\text{TS}}^\ddagger = Q_{\text{TS}}(R_{\text{cm}}^\ddagger, T)$ is the pseudo-partition function of the system at the transition state, the transition-state location $R_{\text{cm}}^\ddagger$ being that value of the reaction coordinate R_{cm} which, at a given T , minimizes $Q_{\text{TS}}(R_{\text{cm}}, T)$. This partition function has its zero of energy on the reaction path where the potential is $V^\ddagger = V(R_{\text{cm}}^\ddagger)$. The form of Q_{TS} will be described in detail below. Q_{react} is the partition function of the reactants; $\sigma/\sigma^\ddagger$ is the ratio of reactant and transition-state symmetry factors and g_e is the ratio of electronic degeneracy factors for the reactants and transition state. The incorporation of large amplitude transition-state motion is through Q_{TS} . In FTST, the overall rotation of the system and the large amplitude transitional modes (as a group) are separable from the small amplitude conserved modes (as a group), *i.e.*

Table 1 Summary of degrees of freedom for various fragment types

case	description	$N_{q_{\text{int}}}$	N_q	N_{cons}
1	atom + lin. – top	1	4	$3N - 8$
2	atom + non-lin. – top	2	5	$3N - 9$
3	lin. – top + lin. – top	3	6	$3N - 10$
4	lin. – top + non-lin. – top	4	7	$3N - 11$
5	non-lin. – top + non-lin. – top	5	8	$3N - 12$

$$Q_{\text{TS}}(R_{\text{cm}}, T) = Q_{\text{c}}(R_{\text{cm}}, T) Q_{\text{tr}}(R_{\text{cm}}, T) \quad (2)$$

and

$$k(T) = g_{\text{e}} \frac{1}{\beta h} \frac{\sigma}{\sigma^{\ddagger}} \frac{Q_{\text{c}}^{\ddagger} Q_{\text{tr}}^{\ddagger}}{Q_{\text{react}}(T)} \exp(-\beta V^{\ddagger}) \quad (3)$$

where Q_{c} and Q_{tr} are the conserved mode and transitional/external rotational mode partition functions, respectively. Q_{c} is evaluated quantum mechanically (almost always as independent harmonic oscillators from normal modes, which are assumed to be orthogonal to the reaction coordinate). However, Q_{tr} is treated in the classical limit, *i.e.*

$$Q_{\text{tr}} = \frac{1}{h^{N_q}} \int \dots \int \exp(-\beta H) d\mathbf{p} d\mathbf{q} \quad (4)$$

where H is the classical Hamiltonian for the N_q transitional/external rotational modes (excluding, of course, the reaction coordinate). The integration variables are the generalized coordinates $\mathbf{q}$ and their conjugate momenta $\mathbf{p}$. H can be written as the sum of a transitional-mode kinetic energy T_{tr} and a potential energy V_{tr} . In describing T_{tr} and V_{tr} , the fragments will be assumed to be rigid bodies whose shapes are those at the given value of R_{cm} along the reaction path. That is to say, at any point on the reaction path, each fragment is to some degree adjusting its optimized internal geometry in response to the presence of the other fragment. When either fragment departs from the reaction path in transitional-mode directions, its internal geometry will be presumed fixed and only its orientation relative to the other fragment will change. This assumption means that each fragment, no matter how complicated, can be treated as either an atom or a linear, spherical, symmetric, or asymmetric top for the purposes of transitional motion. Principal moment of inertia analysis of each fragment at position R_{cm} fixes the locations of the atoms of the fragment with respect to the principal axes. The relative orientation of the fragments to each other, *i.e.* the transitional modes, are then determined only by the angles that govern the relative orientation of the principal axes of each top. When decomposed from the full PES, V_{tr} can ultimately be formulated as a function of only those orientation angles and R_{cm} .

With the above assumptions, large amplitude motion is contained exclusively in Q_{tr} , which can be drastically simplified, as described in detail in part I. The first step in the simplification is to analytically do all the momentum integrals. Since V_{tr} cannot depend on $\mathbf{p}$, eqn. (4) can be rewritten as

$$Q_{\text{tr}} = \frac{1}{h^{N_q}} \int \dots \int \exp(-\beta V_{\text{tr}}) \left(\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp(-\beta T_{\text{tr}}) d\mathbf{p} \right) d\mathbf{q} \quad (5)$$

When the kinetic energy is written as a sum of quadratic terms in the generalized velocities, $\dot{\mathbf{q}} = (\dot{q}_1, \dot{q}_2, \dots)$,

$$2T_{\text{tr}} = \sum_{i,j=1}^{N_q} A_{ij} \dot{q}_i \dot{q}_j = 2T_{\text{tr}} = \dot{\mathbf{q}}^T \cdot \mathbf{A}_0 \dot{\mathbf{q}} \quad (6)$$

the integral over the conjugate momenta in eqn. (5) can be performed without having to express $\mathbf{p}$ explicitly,²⁰ with the result,

$$\begin{aligned} Q_{\text{tr}} &= \frac{1}{h^{N_q}} \int \dots \int \left(\frac{2\pi}{\beta} \right)^{N_q/2} |\mathbf{A}_0|^{1/2} \exp(-\beta V_{\text{tr}}) d\mathbf{q} \\ &= \left(\frac{2\pi}{\beta h^2} \right)^{N_q/2} \int \dots \int |\mathbf{A}_0|^{1/2} \exp(-\beta V_{\text{tr}}) d\mathbf{q} \end{aligned} \quad (7)$$

where $|A_0|$ is the determinant of the matrix whose coefficients are A_{ij} .

The expression for $|A_0|$ depends on the definition of the reaction coordinate, since A_0 describes transitional motion referred to that coordinate. In part I, specific expressions were derived for $|A_0|$ for when R_{cm} is the reaction coordinate. These expressions have a relatively simple form and are given for cases 1–5 in Table 2.

To provide an example of the generalized coordinates, the associated matrix A_0 , and its determinant, we use case 5 as considered in detail in part I, *i.e.* two dissimilar symmetric top fragments with principal moments of inertia (I_{1a}, I_{1a}, I_{1b}) and (I_{2a}, I_{2a}, I_{2b}) . In general, these moments (and hence A_0) are functions of R_{cm} , owing to the dependence of a fragment's equilibrium geometry on the reaction coordinate, but this dependence is suppressed in what follows. Determinants for cases 1–4 can be independently derived but can also be inferred by systematically reducing the symmetric tops of case 5 to linear tops and atoms. The extension of the symmetric-top version of case 5 to dissimilar asymmetric or spherical tops is outlined in part I.

It is useful to divide the kinetic energy for angular motion, T_{tr} , into three terms:²¹

$$\begin{aligned} 2T_{tr} &= 2(T_{tr, \text{int}} + T_{tr, \text{int-ext}} + T_{tr, \text{ext}}) \\ &= \dot{\mathbf{q}}^T \cdot \mathbf{A}_0 \cdot \dot{\mathbf{q}} \end{aligned} \quad (8)$$

The first term in eqn. (8) is the internal kinetic energy, $T_{tr, \text{int}}$, the second term represents coupling between internal and external angular momenta (Coriolis coupling), $T_{tr, \text{int-ext}}$, and the last term is the energy associated with the motion of the body-fixed axes, $T_{tr, \text{ext}}$. For the case of two non-linear tops, $\dot{\mathbf{q}} = (\dot{\Psi}, \dot{\Theta}, \dot{\Phi}, \dot{\phi}_{cm1}, \dot{\theta}_{cm1}, \dot{\phi}_{cm2}, \dot{\theta}_{cm2}, \dot{\chi}_{cm})$ where $(\Psi, \Theta$ and $\Phi)$ are external angles describing the orientation of the entire system relative to space-fixed axes, and $(\phi_{cm1}, \phi_{cm2}, \theta_{cm1}, \theta_{cm2},$ and $\chi_{cm})$ are internal (Jacobi) angles specifying the relative orientation of the two rigid symmetric tops. The elements A_{ij} of A_0 are given in Table 1 of part I. After lengthy algebra (performed *via* the MAPLE algebraic manipulator software package), the determinant $|A_0|$ required in eqn. (7), is

$$|A_0| = (\mu R_{cm}^2)^2 I_{1a}^2 I_{1b} I_{2a}^2 I_{2b} \sin^2 \theta_{cm1} \sin^2 \theta_{cm2} \sin^2 \Theta \quad (9)$$

Note that the R_{cm} dependence of the moments of inertia have been suppressed.

Substitution of eqn. (9) into eqn. (7) followed by analytical integration over $\Theta(0$ to $\pi)$, $\Psi(0$ to $2\pi)$ and $\Phi(0$ to $2\pi)$ yields,

$$Q_{tr}(R_{cm}, T) = \Gamma(R_{cm}, T) Q_{pd}(R_{cm}, T) Q_{fr, 1}(R_{cm}, T) Q_{fr, 2}(R_{cm}, T) \quad (10)$$

where $Q_{pd}(R_{cm}, T) = 2\mu R_{cm}^2 / \beta \hbar^2$ is the partition function for the pseudo-diatomic formed from the centres of mass of the two fragments, $Q_{fr, i}(R_{cm}, T) = (2\pi I_{ib} / \beta \hbar^2)^{1/2} (2I_{ia} / \beta \hbar^2)$ is the partition for the free rotation of symmetric top i , and $\Gamma(R_{cm}, T)$ is a hindering function in the form of the configuration integral:

$$\begin{aligned} \Gamma(R_{cm}, T) &\equiv (2^5 \pi^3)^{-1} \int_0^\pi d\theta_{cm1} \sin \theta_{cm1} \int_0^\pi d\theta_{cm2} \sin \theta_{cm2} \int_0^{2\pi} d\phi_{cm1} \int_0^{2\pi} d\phi_{cm2} \int_0^{2\pi} d\chi_{cm} \\ &\times \exp(-\beta V_{tr}) \end{aligned} \quad (11)$$

eqn. (10) has been derived for two symmetric tops. However, because the modifications to $|A_0|$ are so simple for other tops or atomic fragments, it turns out that the equation is generally valid if $Q_{fr, i}(R_{cm}, T)$ is correctly evaluated. For example, in the case of two dissimilar asymmetric top fragments ($I_{ia} \neq I_{ib} \neq I_{ic}$, $i = 1, 2$), eqn. (10) should be evaluated with $Q_{fr, i}(R_{cm}, T)$ of the form:

$$Q_{fr, i}(R_{cm}, T) = \pi^{1/2} \left(\frac{2I_{ia}}{\beta \hbar^2} \right)^{1/2} \left(\frac{2I_{ib}}{\beta \hbar^2} \right)^{1/2} \left(\frac{2I_{ic}}{\beta \hbar^2} \right)^{1/2} \quad (12)$$

Table 2 Determinants for centres-of-mass reaction coordinate

case	internal angle	moments of inertia		$ A'_0 $
		fragment 1	fragment 2	
1	θ_{cm}	—	I	$(\mu R_{cm}^2)^2 I^2 \sin^2 \theta_{cm}$
2	θ_{cm}, ϕ_{cm}	—	I_a, I_b	$(\mu R_{cm}^2)^2 I_a^2 I_b \sin^2 \theta_{cm}$
3	$\theta_{cm1}, \theta_{cm2}, \chi_{cm}$	I_1	I_2	$(\mu R_{cm}^2)^2 I_1^2 I_2^2 \sin^2 \theta_{cm1} \sin^2 \theta_{cm2}$
4	$\theta_{cm1}, \theta_{cm2}, \phi_{cm2}, \chi_{cm}$	I_1	I_{2a}, I_{2b}	$(\mu R_{cm}^2)^2 I_1^2 I_{2a}^2 I_{2b} \sin^2 \theta_{cm1} \sin^2 \theta_{cm2}$
5	$\theta_{cm1}, \phi_{cm1}, \theta_{cm2}, \phi_{cm2}, \chi_{cm}$	I_{1a}, I_{2b}	I_{2a}, I_{2b}	$(\mu R_{cm}^2)^2 I_{1a}^2 I_{1b} I_{2a}^2 I_{2b} \sin^2 \theta_{cm1} \sin^2 \theta_{cm2}$

This expression of Q_{tr} is exact within the framework of canonical FTST with the centres-of-mass separation as the reaction coordinate. The expression has a particularly appealing form: it is a product of a kinematics factor and a hindering function. The kinematic factor is itself a product of standard rotational partition functions for non-interacting fragments. These partition functions are completely independent of V_{tr} . They depend only on the masses and on the shape changes of the fragments as they progress down the reaction path. The hindering function, eqn. (11), is independent of the masses and is exclusively dependent on V_{tr} . The hindering function is just $\exp(-\beta V_{tr})$ averaged over the (internal) angles required to specify the relative orientation of the two fragments. Note that when $V_{tr} = 0$ (i.e. no inter-fragment interaction), $\Gamma(R_{cm}, T) = 1$. By definition, V_{tr} is zero at the reaction-path geometry. If the reaction-path geometry has the most attractive geometry at a given value of R_{cm} (as is generally the case), then at that value of R_{cm} , V_{tr} will by definition never be less than zero and consequently the hindering function will always fall between 0 and 1. Thus Q_{tr} is generally a potential-dependent fraction times a shape- and mass-dependent kinematic factor.

Minimization of the partition function product $Q_c(R_{cm}, T)Q_{tr}(R_{cm}, T)$ [eqn. (2)] with respect to R_{cm} yields $Q_c^\dagger(T)Q_{tr}^\dagger(T) = Q_c^\dagger(T)\Gamma^\dagger(T)Q_{pd}^\dagger(T)Q_{fr,1}^\dagger(T)Q_{fr,2}^\dagger(T)$. Substitution of the latter expression into the rate-coefficient expression [eqn. (3)] gives;

$$k(T) = g_e \frac{1}{\beta h} \frac{\sigma}{\sigma^\dagger} \frac{Q_c^\dagger(T)\Gamma^\dagger(T)Q_{pd}^\dagger(T)Q_{fr,1}^\dagger(T)Q_{fr,2}^\dagger(T)\exp(-\beta V^\dagger)}{Q_{react}(T)} \quad (13)$$

For an association reaction, $Q_{react} = Q_{trans} Q_{vib,1} Q_{vib,2} Q_{fr,1} Q_{fr,2}$ and $k(T)$ becomes

$$k(T) = g_e \frac{1}{\beta h} \frac{\sigma}{\sigma^\dagger} \frac{\exp(-\beta V^\dagger)}{Q_{trans}(T)} \left[\frac{Q_c^\dagger(T)}{Q_{vib,1}(T)Q_{vib,2}(T)} \right] \left[\frac{Q_{pd}^\dagger(T)Q_{fr,1}^\dagger(T)Q_{fr,2}^\dagger(T)}{Q_{fr,1}(T)Q_{fr,2}(T)} \right] \Gamma^\dagger(T) \quad (14)$$

Of the last three factors in this expression, the first contains all the vibrational information, the second all the shape information, and the last all the transitional-mode potential information.

3. Extension to Arbitrary Reaction Paths

3.1. Discussion of Approach

In part I, we choose the CM separation as the reaction coordinate and locate a particular point on the reaction path by fixing the value of R_{cm} . The relative orientation of the fragments is specified by rotating the fragments about hinge points located at either end of the line connecting the centres of mass of the two fragments. However, a more general approach is to allow for arbitrary hinge points about which to rotate the fragments. The associated reaction coordinate is the line connecting the displaced hinge points; a point on the reaction path is now located by fixing the value of R . Such a generalization is depicted in Fig. 1 for the atom-diatom case (case 1). As a consequence, the distance R_{cm} depends on the internal angles (just θ_{cm} for case 1 in Fig. 1) for a given value of R :

$$R_{cm} = F(\text{internal angles}; R) \quad (15)$$

where F is some completely flexible function capable of representing all ways that R_{cm} could depend on angles while holding a separation distance R fixed. Explicit expressions for this general constraint of R_{cm} will be given in the next section for a simple case. This situation implies that when time derivatives of cartesian coordinates are taken, as is required in the construction of the kinetic-energy matrix, R_{cm} can in general have a non-zero time derivative since it is now a function of the internal angles which do have a non-zero time derivative. It is only through such derivatives that new terms arise in the kinetic-energy matrix. These new terms will affect the determinant. Note that because

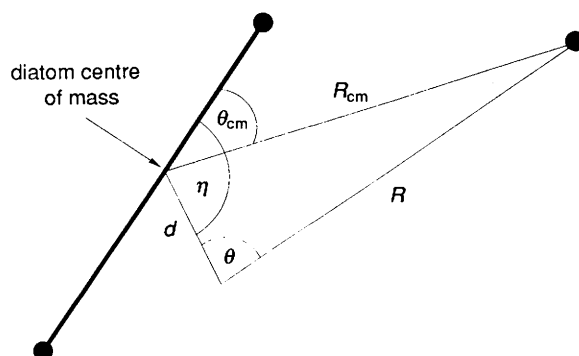


Fig. 1 Schematic representation of the geometric constraints on R_{cm} for a fixed separation R and a completely arbitrary reaction coordinate parameterized by d and η

the new terms enter only through the time derivatives of internal angles, they only appear in that part of the matrix connected to the coriolis and internal kinetic-energy terms. These new terms become kinetically interesting when they lower the value of the determinant (at least some values of internal angles), for then there is at least a chance that the rate constant can be further minimized. It will be shown in the next section that it is possible to select a coordinate system such that these additional terms arise only in the internal kinetic-energy part of the matrix and never in the coriolis part of the matrix. The objectives of this section are: (a) to express a completely general F function; (b) to select an appropriate coordinate system in which these additional terms only appear in the internal kinetic energy; (c) to establish the form of the modified A matrix whose determinant we seek.

3.2. General Formulae

The basic formula for the kinetic energy is

$$\begin{aligned}
 2T_{tr} &= 2(T_{tr, \text{int}} + T_{tr, \text{int-ext}} + T_{tr, \text{ext}}) \\
 &= \sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i + 2\boldsymbol{\omega} \cdot \sum_{i=1}^N m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i + \boldsymbol{\omega}^T \cdot \mathbf{I} \cdot \boldsymbol{\omega} \\
 &= (\dot{\mathbf{q}}_{\text{ext}}, \dot{\mathbf{q}}_{\text{int}})^T \cdot \mathbf{A} \cdot (\dot{\mathbf{q}}_{\text{ext}}, \dot{\mathbf{q}}_{\text{int}}) \\
 &= (\boldsymbol{\omega}, \dot{\mathbf{q}}_{\text{int}})^T \cdot \mathbf{A}' \cdot (\boldsymbol{\omega}, \dot{\mathbf{q}}_{\text{int}})
 \end{aligned} \tag{16}$$

where $\boldsymbol{\omega}$ is the angular velocity vector in the internal coordinate system, $\mathbf{r}_i$ is the cartesian vector for each particle of mass m_i and $\dot{\mathbf{r}}_i$ is the time derivative of $\mathbf{r}_i$.

$\mathbf{A}$ is the matrix whose determinant we seek but it is simpler to work with $\mathbf{A}'$ which is related to $\mathbf{A}$ by a similarity transformation involving the time derivatives of $\mathbf{q}_{\text{ext}} = (\Psi, \Theta, \Phi)$. This transformation matrix connecting $\boldsymbol{\omega}$ and $\dot{\mathbf{q}}_{\text{ext}}$ is derived in Ref. 22. Since determinants of similarity transforms are products of the determinants of the original matrix, the transformation matrix, and its inverse, we have $|\mathbf{A}| = \sin^2 \Theta |\mathbf{A}'|$.

The formula for $2T_{tr}$ is based on the notion that rigid body kinematics can be represented by motion of a fixed point attached to the body plus rotation about the fixed point. The most convenient selection of an internal coordinate system is one that has its origin at this fixed point. The actual fixed point selected will change $\mathbf{A}'$ and the determinant. It can be shown that the best fixed point is the collision system's centre of mass. Any other fixed point will increase $|\mathbf{A}'|$ for all angles and thus is not kinetically interesting.

With the fixed point selected, the orientation of the axes at the fixed point is completely optional. A similarity transform that reorients the axes *via* orthogonal transformations leaves the determinant of the transformed A' unchanged from that of the original A' . Thus, as in part I, we choose R_{cm} as the z axis of the coordinate system. This has a number of desirable qualities which are quite relevant to the question of including the F function constrain on R_{cm} .

We write A' in a very general form. To begin, we define the following quantities:

$$H_{xx} = \sum_{i=1}^N m_i x_i^2, H_{xy} = \sum_{i=1}^N m_i x_i y_i, \dots \quad (17)$$

$$H_{xxj} = \sum_{i=1}^N m_i x_i (dx_i/dq_j), H_{xyj} = \sum_{i=1}^N m_i x_i (dy_i/dq_j), \dots \quad (18)$$

$$H_{xjxk} = \sum_{i=1}^N m_i (dx_i/dq_j)(dx_i/dq_k), H_{xjyk} = \sum_{i=1}^N m_i (dx_i/dq_j)(dy_i/dq_k), \dots \quad (19)$$

$$A' = \begin{bmatrix} A_{rot} & A_{cor} \\ A_{cor}^T & A_{int} \end{bmatrix} \quad (20)$$

Here $H_{xx}, H_{xy} \dots$ are related to the usual moment of inertia expressions (*e.g.* $I_{xx} = H_{yy} + H_{zz}$; $I_{xy} = -H_{xy}$), whereas H_{xxj}, H_{xjxk} , etc. are derivatives of the H s with respect to internal coordinates. Here A_{rot} is that part of A' that is multiplied by products of ω components. It comes from the third term for rigid rotation in the basic formula [eqn. (16)] and for all cases is a 3×3 matrix. A_{cor} is that part of A' that involves cross products of ω components and q_{int} components. It comes from the second term for Coriolis coupling in the basic formula and is a rectangular $N_{q_{int}} \times 3$ matrix, where $N_{q_{int}}$ is the number of internal angles which varies from 1 to 5, as indicated in Table 1. A_{int} is that part of A' that involves products of q_{int} components. It comes from the first term in the basic formula and is an $N_{q_{int}} \times N_{q_{int}}$ matrix.

With the above definitions,

$$\sum_{i=1}^N m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i = \begin{bmatrix} H_{yz1} - H_{zy1} & H_{yz2} - H_{zy2} & \dots \\ H_{zx1} - H_{xz1} & H_{zx2} - H_{xz2} & \dots \\ H_{xy1} - H_{yx1} & H_{xy2} - H_{yx2} & \dots \end{bmatrix} \begin{bmatrix} \dot{q}_{int,1} \\ \dot{q}_{int,2} \\ \vdots \end{bmatrix} \quad (21)$$

$$\sum_{i=1}^N m_i \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i = [\dot{q}_{int,1} \quad \dot{q}_{int,2} \quad \dots] \times \begin{bmatrix} H_{x1x1} + H_{y1y1} + H_{z1z1} & H_{x1x2} + H_{y1y2} + H_{z1z2} & \dots \\ H_{x2x1} + H_{y2y1} + H_{z2z1} & H_{x2x2} + H_{y2y2} + H_{z2z2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} \dot{q}_{int,1} \\ \dot{q}_{int,2} \\ \vdots \end{bmatrix} \quad (22)$$

$$[\omega, q_{int}] \cdot A' \cdot \begin{bmatrix} \omega \\ q_{int} \end{bmatrix} = \omega \cdot A_{rot} \cdot \omega + \dot{q}_{int}^T \cdot A_{int} \cdot \dot{q}_{int} + \omega^T \cdot A_{cor} \cdot \dot{q}_{int} + \dot{q}_{int}^T \cdot A_{cor} \cdot \omega \quad (23)$$

Based on eqn. (16), (21), (22), and (23), the following identifications are readily made:

$$A_{rot} = I \quad (24)$$

$$A_{\text{int}} = \begin{bmatrix} H_{x1x1} + H_{y1y1} + H_{z1z1} & H_{x1x2} + H_{y1y2} + H_{z1z2} & \cdots \\ H_{x2x1} + H_{y2y1} + H_{z2z1} & H_{x2x2} + H_{y2y2} + H_{z2z2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (25)$$

$$A_{\text{cor}} = \begin{bmatrix} H_{yz1} - H_{zyl} & H_{yz2} - H_{zy2} & \cdots \\ H_{zx1} - H_{xz1} & H_{zx2} - H_{xz2} & \cdots \\ H_{xy1} - H_{yx1} & H_{xy2} - H_{yx2} & \cdots \end{bmatrix} \quad (26)$$

These formulae have already been implicitly used in part I for all five cases in order to get the determinants for a centres-of-mass reaction coordinate.

Now we want to take explicit account of eqn. (15) and the details of the coordinate system to determine how A_{rot} , A_{int} and A_{cor} change. To begin with, every place where R_{cm} appears in A_{rot} , A_{int} and A_{cor} requires a substitution of the F function from eqn. (15). However, for our purposes, we do not have to do that before deriving the determinant. One can make this substitution in the determinant directly. Thus everywhere R_{cm} appears in $|A'|$, we can substitute an expression, as yet unspecified, of angles and the separation R .

Before constructing the determinant, it is necessary to develop the new terms that come from the fact that R_{cm} is now an explicit function of the internal angles. These new terms only arise through the derivative expressions involving H_{xij} , etc. This can be illustrated by analysing H_{yx1} . This has the form:

$$H_{yx1} = \sum_{i=1}^N m_i y_i (dx_i/dq_{\text{int},1}) = \sum_{i=1}^N m_i y_i [(\partial x_i/\partial q_{\text{int},1}) + (\partial x_i/\partial R_{\text{cm}})(\partial R_{\text{cm}}/\partial q_{\text{int},1})] \quad (27)$$

where $(\partial x_i/\partial q_{\text{int},1})$ indicates the explicit dependence of x_i on $q_{\text{int},1}$ (with R_{cm} and all other q_{int} s held constant) and $(\partial x_i/\partial R_{\text{cm}})(\partial R_{\text{cm}}/\partial q_{\text{int},1})$ indicates the implicit dependence through the dependence of R_{cm} on $q_{\text{int},1}$ via eqn. (15) (with all q_{int} s held constant in the first factor and with R and all other q_{int} s held constant in the second). This second term does not exist in part I because the F in eqn. (15) has always been treated as a constant, making $\partial R_{\text{cm}}/\partial q_{\text{int},1}$ zero. Now we consider it not to be zero.

Even though eqn. (15) is a very general constraint, making $(\partial R_{\text{cm}}/\partial q_{\text{int},1})$ of as yet undetermined complexity, the specific selection of the coordinate system has implications for the terms $(\partial v_i/\partial R_{\text{cm}})$ where $v = x, y$, or z . In particular, since R_{cm} is the z axis of the coordinate system,

$$(\partial x_i/\partial R_{\text{cm}}) = (\partial y_i/\partial R_{\text{cm}}) = 0 \quad (28)$$

because the common x and y axes for both reactants are perpendicular to the z axis. Furthermore, changing R_{cm} only causes a common displacement for every atom on one reactant and a different but common displacement for every atom on the second reactant. Consequently, if u stands for either x or y :

$$\begin{aligned} H_{uzj} &= \sum_{i=1}^N m_i u_i [(\partial z_i/\partial q_{\text{int},j}) + (\partial z_i/\partial R_{\text{cm}})(\partial R_{\text{cm}}/\partial q_{\text{int},j})] \\ &= \sum_{i=1}^N m_i u_i \left\{ (\partial z_i/\partial q_{\text{int},j}) \right. \\ &\quad \left. + (\partial R_{\text{cm}}/\partial q_{\text{int},j}) \left[\sum_{i_1=1}^{N_1} m_{i_1} u_{i_1} (\partial z_{i_1}/\partial R_{\text{cm}}) + \sum_{i_2=1}^{N_2} m_{i_2} u_{i_2} (\partial z_{i_2}/\partial R_{\text{cm}}) \right] \right\} \end{aligned} \quad (29)$$

where i_1 and i_2 label atoms in fragments 1 and 2, respectively, and $N_1 + N_2 = N$. Since $(\partial z_{i_1}/\partial R_{\text{cm}})$ and $(\partial z_{i_2}/\partial R_{\text{cm}})$ are constant within each fragment, they can be set equal to

Δz_1 and Δz_2 , respectively. Now eqn. (29) becomes

$$H_{uzj} = \sum_{i=1}^N m_i u_i \left\{ (\partial z_i / \partial q_{\text{int}, j}) + (\partial R_{\text{cm}} / \partial q_{\text{int}, j}) \left[\Delta z_1 \sum_{i_1}^{N_1} m_{i_1} u_{i_1} + \Delta z_2 \sum_{i_2}^{N_2} m_{i_2} u_{i_2} \right] \right\} \\ = \sum_{i=1}^N m_i u_i (\partial z_i / \partial q_{\text{int}, j}) \quad (30)$$

where the fact that R_{cm} connects centres of mass for each reactant guarantees that $\sum m_i u_i = 0$ when the sum is restricted to either reactant and $u = x$ or y . Eqn. (30) is the expression we implicitly use in part I and thus the constraint of eqn. (15) in this coordinate system has no effect on any H_{uvj} , where u and v are x , y and z .

The above arguments imply that in the selected coordinate system, no new terms arising from eqn. (15) develop in A_{rot} and A_{cor} . Given the nature of the terms in the remaining component of A' , namely A_{int} , the only new terms that occur are, *via* H_{zjk} :

$$H_{zjk} = \sum_{i=1}^N m_i [(\partial z_i / \partial q_{\text{int}, j}) \\ + (\partial z_i / \partial R_{\text{cm}})(\partial R_{\text{cm}} / \partial q_{\text{int}, j})][(\partial z_j / \partial q_{\text{int}, k}) + (\partial z_j / \partial R_{\text{cm}})(\partial R_{\text{cm}} / \partial q_{\text{int}, k})] \\ = \sum_{i=1}^N m_i (\partial z_i / \partial q_{\text{int}, j})(\partial z_i / \partial q_{\text{int}, k}) \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, j}) \left[\sum_{i_1}^{N_1} m_{i_1} (\partial z_{i_1} / \partial q_{\text{int}, j})(\partial z_{i_1} / \partial R_{\text{cm}}) + \sum_{i_2}^{N_2} m_{i_2} (\partial z_{i_2} / \partial q_{\text{int}, j})(\partial z_{i_2} / \partial R_{\text{cm}}) \right] \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, k}) \left[\sum_{i_1}^{N_1} m_{i_1} (\partial z_{i_1} / \partial q_{\text{int}, k})(\partial z_{i_1} / \partial R_{\text{cm}}) + \sum_{i_2}^{N_2} m_{i_2} (\partial z_{i_2} / \partial q_{\text{int}, k})(\partial z_{i_2} / \partial R_{\text{cm}}) \right] \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, j})(\partial R_{\text{cm}} / \partial q_{\text{int}, k}) \sum_i^n m_i (\partial z_i / \partial R_{\text{cm}})^2 \quad (31) \\ = \sum_{i=1}^N m_i (\partial z_i / \partial q_{\text{int}, j})(\partial z_i / \partial q_{\text{int}, k}) \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, j}) \left[\Delta z_1 \sum_{i_1}^{N_1} m_{i_1} (\partial z_{i_1} / \partial q_{\text{int}, j}) + \Delta z_2 \sum_{i_2}^{N_2} m_{i_2} (\partial z_{i_2} / \partial q_{\text{int}, j}) \right] \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, k}) \left[\Delta z_1 \sum_{i_1}^{N_1} m_{i_1} (\partial z_{i_1} / \partial q_{\text{int}, k}) + \Delta z_2 \sum_{i_2}^{N_2} m_{i_2} (\partial z_{i_2} / \partial q_{\text{int}, k}) \right] \\ + (\partial R_{\text{cm}} / \partial q_{\text{int}, j})(\partial R_{\text{cm}} / \partial q_{\text{int}, k}) \sum_i^n m_i (\partial z_i / \partial R_{\text{cm}})^2 \quad (32)$$

where $(\partial z_{i_1} / \partial R_{\text{cm}})$ and $(\partial z_{i_2} / \partial R_{\text{cm}})$ have been replaced by Δz_1 and Δz_2 in going from eqn. (31) to eqn. (32). To proceed further, we note that in our coordinate system:

$$z_{i_1} = -f R_{\text{cm}} + z'_{i_1} \\ z_{i_2} = (1 - f) R_{\text{cm}} + z'_{i_2} \quad (33) \\ f = \sum_{i_2}^{N_2} m_{i_2} / \sum_{i=1}^N m_i = m_2 / M$$

where z' is measured in the local fragment centres-of-mass coordinate system such that

$$\sum_{i_1}^{N_1} m_{i_1} z'_{i_1} = \sum_{i_2}^{N_2} m_{i_2} z'_{i_2} = 0 \quad (34)$$

This equation makes the middle two terms in eqn. (32) become zero. The last term of eqn. (32) assumes the form:

$$(\partial R_{\text{cm}}/\partial q_{\text{int},j})(\partial R_{\text{cm}}/\partial q_{\text{int},k}) \left[f^2 \sum_{i_1}^{N_1} m_{i_1} + (1-f)^2 \sum_{i_2}^{N_2} m_{i_2} \right] = \mu (\partial R_{\text{cm}}/\partial q_{\text{int},j})(\partial R_{\text{cm}}/\partial q_{\text{int},k}) \quad (35)$$

where $\mu = m_1 m_2 / M$ is the reduced mass of fragments 1 and 2.

If we define A'_0 as the form of A' when the F function is just a constant and if we define the vector ∇F by:

$$\nabla F = \begin{bmatrix} \partial F/\partial q_{\text{int},1} \\ \partial F/\partial q_{\text{int},2} \\ \vdots \\ \vdots \end{bmatrix} \quad (36)$$

then the A' matrix is related to the A'_0 matrix by

$$A' = A'_0(R_{\text{cm}} = F) + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0}^T & \mu \nabla F \nabla F^T \end{bmatrix} \quad (37)$$

where $\mathbf{1}$ is the 3×3 identity matrix, $\mathbf{0}$ is the $3 \times N_q$ null matrix, and $A'_0(R_{\text{cm}} = F)$ means the old matrix where F is substituted for R_{cm} . Thus, if we can write down a general form for the constraint F , we need only evaluate F , add it to A'_0 , form the determinant, and then substitute F for R_{cm} . Since we have already the form of A'_0 in the coordinate system used here, all we need to evaluate a completely general determinant is an expression for F of eqn. (15).

4. Case-specific Formulae

The constraint F varies from case to case, but a specific expression for each of cases 1 through 5 can be formulated without undue difficulty. Since both the dimension of the original A'_0 matrix and the complexity of the modification embodied in $\mu \nabla F \nabla F^T$ increases from case 1 to case 5, determination of a compact analytical form for the determinant of the new A' matrix [eqn. (37)] by hand is generally not feasible. As was done in part I for $|A'_0|$, it is expedient to employ an algebraic manipulator to obtain $|A'|$. Even so, the resulting analytical expressions for $|A'|$ are not as compact or physically transparent as for $|A'_0|$ (see Table 2 for the latter). This is not, however, a matter of great concern: The $|A'|$ expressions (i) are nevertheless analytical, which is a significant improvement over any existing variable reaction coordinate treatments in the broad context of VTST; and (ii) will ultimately appear in a configuration integral [analogous to eqn. (7)] which will be evaluated numerically. It is nevertheless of pedagogical importance to illustrate the implementation of the method derived in Section 3. We choose the simplest possible case, case 1 for an atom-diatom system, since there is only one transitional degree of freedom and since all algebraic manipulations are readily performed by hand and the ensuing expressions are not too cumbersome.

The various distances and angles involved in specifying an arbitrary reaction coordinate for the atom-diatom case are indicated in Fig. 1. $(R_{\text{cm}}, \theta_{\text{cm}})$ denote the centres-of-mass reaction coordinate and the (Jacobi) internal angle associated with the single transitional degree of freedom, respectively. The generalized reaction coordinate and its associated internal angle are designated (R, θ) . The reaction coordinate is general, in that the hinge point at the diatomic end of R is at the tip of the vector d whose magnitude d and orientation η with respect to the diatomic axis are arbitrary. A particular reaction coordinate is selected by fixing d and η . The associated transitional

mode now consists of all possible rotations of $\mathbf{d}$ about the displaced hinge point; a particular point on the reaction coordinate is now specified by fixing R .

Here we confine $\mathbf{d}$ to the triatomic plane but, at least in principle, $\mathbf{d}$ can lie out of this plane; one needs to specify only an azimuthal angle in addition to the polar angle η . The out-of-plane hinge point is not further considered because its physical significance for an atom–diatom case is not apparent and because it unnecessarily complicates our example without providing additional insight into the essential features. The out-of-plane case has obvious physical relevance for the non-linear top–atom case (case 2), whose treatment we defer to a subsequent paper.

From Fig. 1 it can be seen that R_{cm} depends on θ or, equivalently, on $\lambda = \eta - \theta_{\text{cm}}$ for fixed R . Choosing the latter as the independent internal angle variable is consistent with the approach described in Section 3. The relationship between R , R_{cm} and θ_{cm} is simply expressed using the law of cosines:

$$R^2 = R_{\text{cm}}^2 + d^2 - 2dR_{\text{cm}} \cos \lambda \quad (38)$$

or

$$R_{\text{cm}} = d \cos \lambda + [R^2 - d^2 \sin^2 \lambda]^{1/2} = F(\theta_{\text{cm}}; R) \quad (39)$$

We are now in a position to derive the determinant expression for an arbitrary reaction coordinate for the atom–diatom case by modifying $\mathbf{A}'_0$ derived in part I by using the above F function in eqn. (37). In this case $\mathbf{A}'_0$ is a 4×4 matrix since there are three external rotational degrees of freedom and one internal (transitional) degree of freedom. With the z axis of the coordinate system pointing along R_{cm} , and with the y axis in the triatomic plane, it has the following components:

$$\mathbf{A}'_0 = \begin{bmatrix} I + \mu R_{\text{cm}}^2 & 0 & 0 & I \\ 0 & \mu R_{\text{cm}}^2 + I \cos^2 \theta_{\text{cm}} & -I \cos \theta_{\text{cm}} \sin \theta_{\text{cm}} & 0 \\ 0 & -I \cos \theta_{\text{cm}} \sin \theta_{\text{cm}} & I \sin^2 \theta_{\text{cm}} & 0 \\ I & 0 & 0 & I \end{bmatrix} \quad (40)$$

The kinetic energy, T_{tr} , is given by eqn. (8) with $\mathbf{q} = (\omega, \mathbf{q}_{\text{int}}) = (\omega_x, \omega_y, \omega_x, \theta_{\text{cm}})$ and $\mathbf{A}'_0$ by eqn. (40). The determinant, $|\mathbf{A}'_0|$, for a centres-of-mass reaction coordinate for case 1 is given in Table 2. Next one needs ∇F , which in this case has the single element of $\partial R_{\text{cm}}/\partial \theta_{\text{cm}}$. From eqn. (39), this is

$$\begin{aligned} \nabla F &= \partial R_{\text{cm}}/\partial \theta_{\text{cm}} = -\partial R_{\text{cm}}/\partial \lambda \\ &= d \sin \lambda + d^2 \sin \lambda \cos \lambda / [R_{\text{cm}} - d \cos \lambda] \\ &= R_{\text{cm}} d \sin \lambda / [R_{\text{cm}} - d \cos \lambda] \end{aligned} \quad (41)$$

To evaluate eqn. (37), we will need the square of this expression:

$$(\nabla F)^2 = R_{\text{cm}}^2 d^2 \sin^2 \lambda / [R_{\text{cm}} - d \cos \lambda]^2 \quad (42)$$

Thus from eqn. (37), the final general expression is

$$\mathbf{A}' = \begin{bmatrix} I + \mu R_{\text{cm}}^2 & 0 & 0 & I \\ 0 & \mu R_{\text{cm}}^2 + I \cos^2 \theta_{\text{cm}} & -I \cos \theta_{\text{cm}} \sin \theta_{\text{cm}} & 0 \\ 0 & -I \cos \theta_{\text{cm}} \sin \theta_{\text{cm}} & I \sin^2 \theta_{\text{cm}} & 0 \\ I & 0 & 0 & I + \frac{\mu R_{\text{cm}}^2 d \sin^2 \lambda}{[R_{\text{cm}} - d \cos \lambda]^2} \end{bmatrix} \quad (43)$$

Here each R_{cm} could be replaced by eqn. (39) but it would be simpler to form the determinant and then to make the replacement. Accordingly,

$$|A'| = I^2 \sin^2 \theta_{\text{cm}} (\mu R_{\text{cm}}^2)^2 \left(\frac{R_{\text{cm}}^2 + d^2 - 2dR_{\text{cm}} \cos \lambda + (\mu R_{\text{cm}}^2/I)d^2 \sin^2 \lambda}{[R_{\text{cm}} - d \cos \lambda]^2} \right) \quad (44)$$

$$= I^2 \sin^2 \theta_{\text{cm}} [\mu F^2(\theta_{\text{cm}}; R)]^2 \left[\frac{R^2 + F^2(\theta_{\text{cm}}; R)(\mu d^2/I) \sin^2(\eta - \theta_{\text{cm}})}{R^2 - d^2 \sin^2(\eta - \theta_{\text{cm}})} \right] \quad (45)$$

where $F(\theta_{\text{cm}}; R)$ is given by eqn. (39). Implications of this result are investigated in the next section.

5. Discussion

In order to assess the conceptual and computational implications of an arbitrary reaction coordinate in the context of CFTST, it is useful to compare with the centres-of-mass reaction coordinate results derived in part I and summarized in Section 2. To make a specific comparison, we use the atom-diatom system of Section 4 and consider the transitional/external-rotational partition function, Q_{tr} . For a centres-of-mass reaction coordinate with $N_q = 4$ and $A_0 = \sin^2 \Theta A'_0$ with A'_0 from case 1 in Table 2.

$$\begin{aligned} Q_{\text{tr}}(R_{\text{cm}}, T) &= \left(\frac{2\pi}{\beta h^2} \right)^{N_q/2} \int \cdots \int d\mathbf{q} |A_0|^{1/2} \exp(-\beta V_{\text{tr}}) \\ &= \left(\frac{2\pi}{\beta h^2} \right)^2 \int \cdots \int d(\Psi, \Theta, \Phi, \theta_{\text{cm}}) [\sin^2 \Theta |A'_0(\theta_{\text{cm}}, R_{\text{cm}})|]^{1/2} \exp(-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})) \\ &= \left(\frac{2\pi}{\beta h^2} \right)^2 \int_0^{2\pi} d\Psi \int_0^\pi d\Theta \sin \Theta \int_0^{2\pi} d\Phi \int_0^\pi d\theta_{\text{cm}} |A'_0(\theta_{\text{cm}}, R_{\text{cm}})|^{1/2} \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \\ &= \left(\frac{2\pi}{\beta h^2} \right)^2 8\pi^2 \int_0^\pi d\theta_{\text{cm}} |A'_0(\theta_{\text{cm}}, R_{\text{cm}})|^{1/2} \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \quad (46) \end{aligned}$$

$$\begin{aligned} &= \left(\frac{2\pi}{\beta h^2} \right)^2 8\pi^2 \int_0^\pi d\theta_{\text{cm}} \sin \theta_{\text{cm}} I(\mu R_{\text{cm}}^2) \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \\ &= \left(\frac{2I}{\beta h^2} \right) \left(\frac{2\mu R_{\text{cm}}^2}{\beta h^2} \right) \frac{1}{2} \int_0^\pi d\theta_{\text{cm}} \sin \theta_{\text{cm}} \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \quad (47) \end{aligned}$$

$$= Q_{\text{fr}}(R_{\text{cm}}, T) Q_{\text{pd}}(R_{\text{cm}}, T) \Gamma(R_{\text{cm}}, T) \quad (48)$$

In eqn. (48) a kinematic factor consisting of the free rotor and pseudo-diatomic rotor partition functions $Q_{\text{fr}} = (2I/\beta h^2)$ and $Q_{\text{pd}} = (2\mu R_{\text{cm}}^2/\beta h^2)$ can be identified, and a hindering function Γ in the form of a configurational integral has been defined:

$$\Gamma(R_{\text{cm}}, T) = \frac{1}{2} \int_0^\pi d\theta_{\text{cm}} \sin \theta_{\text{cm}} \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \quad (49)$$

The kinematic factor is a product of standard rotational partition functions for non-interacting fragments, depends only on the masses and the geometry, and is completely independent of V_{tr} . The hindering function, on the other hand, depends exclusively on V_{tr} , $\exp(-\beta V_{\text{tr}})$ being averaged over the internal angle θ_{cm} weighted by $\sin \theta_{\text{cm}}$.

The corresponding partition function expression for an arbitrary reaction coordinate for the atom–diatom system is obtained by substituting A' [eqn. (45)] for A'_0 in eqn. (46):

$$Q_{\text{tr}}(R, d, \eta, T) = \left(\frac{2I}{\beta \hbar^2} \right) \left(\frac{2\mu}{\beta \hbar^2} \right) \frac{1}{2} \int_0^\pi d\theta_{\text{cm}} \sin \theta_{\text{cm}} \\ \times F^2(\theta_{\text{cm}}; R) \left(\frac{R^2 + F^2(\theta_{\text{cm}}; R)(\mu d^2/I) \sin^2(\eta - \theta_{\text{cm}})}{R^2 - d^2 \sin^2(\eta - \theta_{\text{cm}})} \right)^{1/2} \exp[-\beta V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})] \quad (50)$$

with $F(\theta_{\text{cm}}; R) = d \cos(\eta - \theta_{\text{cm}}) + [R^2 - d^2 \sin^2(\eta - \theta_{\text{cm}})]^{1/2}$. It is easy to verify that eqn. (50) reduces to eqn. (47) when $d \rightarrow 0$ (and hence $R \rightarrow R_{\text{cm}}$). The complete factorization of Q_{tr} observed for the centres-of-mass reaction coordinate and expressed by eqn. (48) is not achieved for an arbitrary reaction coordinate. The hindering function now depends on masses and geometry *via* the ratio $\mu d^2/I$. The kinematic factor remains just that: it still depends only on masses and geometry but no longer contains the pseudo-diatomic partition function, Q_{pd} . Nevertheless, the numerical evaluation of the configuration integral in eqn. (50) is not more difficult or time consuming than the original configuration integral in eqn. (49) and the cost of evaluating the latter is trivial no matter what the nature of the potential V_{tr} . The same quadrature method and the same grid size can be used. The same form of the potential, namely $V_{\text{tr}} = V_{\text{tr}}(\theta_{\text{cm}}, R_{\text{cm}})$, appears in both configuration integrals. Thus any coordinate transformation (*e.g.* from centres-of-mass coordinates to bond coordinates) required to evaluate the potential must be applied to obtain both integrands. The difference between these two configuration integrals is the relative weighting of different regions of the θ_{cm} angular space due to the factor

$$F^2(\theta_{\text{cm}}; R)^2 \left(\frac{R^2 + F^2(\theta_{\text{cm}}; R)(\mu d^2/I) \sin^2(\eta - \theta_{\text{cm}})}{R^2 - d^2 \sin^2(\eta - \theta_{\text{cm}})} \right)^{1/2} \quad (51)$$

in the integrand of eqn. (50). This circumstance makes it possible that

$$Q_{\text{TS}}(R, d, \eta, T) = Q_{\text{c}}(R, d, \eta, T) Q_{\text{tr}}(R, d, \eta, T) < Q_{\text{TS}}(R_{\text{cm}}, T) \\ = Q_{\text{c}}(R_{\text{cm}}, T) Q_{\text{tr}}(R_{\text{cm}}, T) \quad (52)$$

for a range of separations S of fragments (*i.e.* $S = R = R_{\text{cm}}$) and for a range of (d, η) values. If this range includes R_{cm}^* , the value of R_{cm} that minimizes Q_{tr} , then a better thermal rate constant prediction emerges. We note that evaluation of Q_{c} relies on knowledge of the frequencies of the conserved modes orthogonal to the reaction coordinate. How to identify these modes for arbitrary reaction coordinates and how to obtain reliable estimates of their frequencies is a matter for further study and is beyond the scope of this paper. Improved canonical rate constants can be obtained by applying the variational criterion to the transition-state partition function, Q_{TS} , as the position along the reaction coordinate and the definition of the reaction coordinate are varied. The resulting reaction path can be said to be optimized in that the definition of the reaction coordinate (*i.e.* the value of d) is continuously varied as the fragments approach or separate. There is a hierarchy of such paths based on the constraints placed on the magnitude d and orientation η of the vector d , whose tip specifies the hinge point for relative orientation of the diatomic fragment: (i) both d and η vary; (ii) one is fixed while the other varies; (iii) both are fixed. We consider the three scenarios in turn.

(i) Fully Optimized Reaction Path Definition

This treatment involves evaluating $Q_{\text{tr}}(R, d, \eta, T)$ [eqn. (50)] on a 3D (R, d, η) grid, finding the minimum value of Q_{TS} [eqn. (52)] on the grid, identifying this minimum as

the transition state value $Q_{\text{TS}}^{\dagger}$, and using the latter in the CFTST rate constant expression [eqn. (1)]. Instead of the transition state being characterized only by a separation $R^{\dagger}$, it is now characterized by a location $(R^{\dagger}, d^{\dagger}, \eta^{\dagger})$ in the space of all possible reaction coordinates and displacements along these coordinates. Implementation of a fully optimized reaction path treatment has, to the best of our knowledge, not yet been undertaken at the microcanonical or canonical levels, even for the atom–diatom system. The method proposed herein makes such treatment feasible. Being at the canonical level, only one minimization is required at each temperature. For a particular choice of reaction coordinate (this includes the centres-of-mass reaction coordinate as a special case), one evaluates eqn. (50) on a 1D grid of n_R values of R . For a fully optimized treatment, the workload goes up by a factor of $n_d \times n_{\eta}$ where n_d and n_{η} are the sizes of the d and η grids. This factor would probably be of the order of 10^2 to 10^3 , which is substantial but not a cause for pessimism because each evaluation of the configuration integral in eqn. (50) is computationally trivial.

(ii) Partly Optimized Reaction Path Definition

Here $Q_{\text{tr}}(R, d, \eta, T)$ is evaluated, and the minimum value of Q_{TS} determined, on a 2D grid of (R, d) with η fixed or (R, η) with d fixed. Connection with the work of Klippenstein is made if we choose the former with η fixed at a value of 0 or π . That is, the diatomic hinge point is displaced a distance d from the diatomic centre-of-mass along the diatomic's symmetry axis. Klippenstein has applied this level of reaction path optimization to $\text{Li} + \text{HF} \rightarrow \text{LiF} + \text{H}$ ^{11a} and to $\text{He} + \text{H}_2^+ \rightarrow \text{HeH}^+ + \text{H}$ ^{11b} using his implementation of microcanonical FTST. He finds that, on significant portions of the ranges of energy and total angular momentum considered, the transition-state 'location' $(R^{\dagger}, d^{\dagger})$ is not on the centres-of-mass reaction coordinate (i.e. $d^{\dagger} \neq 0$), and sometimes lies outside the diatom (i.e. $d^{\dagger} > m_2 r / (m_1 + m_2)$ or $m_1 r / (m_1 + m_2)$, where r is the diatomic bond length). Klippenstein has applied the same level of reaction path optimization to $\text{NCNO} \rightarrow \text{CN} + \text{NO}$ ^{8a} with similar results. In this case, the hinge point in diatoms 1 and 2 is allowed to be displaced by distances d_1 and d_2 along the corresponding diatom's symmetry axis. This situation corresponds to our case 3 (see Table 1) with η_1 and η_2 fixed at 0 or π .

(iii) Fixed Reaction Path Definition

Fixing $d (\neq 0)$ and η specifies a particular reaction coordinate and one that is different from the centres-of-mass reaction coordinate. There is an infinite number of such choices. A particularly interesting choice is a bond reaction coordinate, obtained by specifying $(d = m_2 r / (m_1 + m_2), \eta = \pi)$ or $(d = m_1 r / (m_1 + m_2), \eta = 0)$. Minimizing Q_{TS} with respect to R , which is now an atom–atom separation. Klippenstein has used a bond reaction coordinate at the microcanonical level of FTST for $\text{NCNO} \rightarrow \text{CN} + \text{NO}$ ^{7b} and $\text{CH}_2\text{CO} \rightarrow \text{CH}_2 + \text{CO}$.¹⁰

6. Summary

This paper represents a major step toward our overall objective of an improved and rigorous version of canonical flexible transition-state theory, CFTST. By rigorous, we mean that no new approximations or assumptions are introduced beyond those contained in the original FTST of Wardlaw and Marcus. By improved, we mean conceptual and computational advantages over the original FTST. Both advantages stem from the fact that, at the canonical level of FTST, the multidimensional integral over the phase

space of the transitional/external rotational modes analytically simplifies to an expression containing a configuration integral only over positional coordinates involved in large amplitude (transitional) motion. These coordinates are internal (Jacobi) angles in terms of which the transitional-mode potential energy, V_{tr} , may be directly expressed or which are easily related to the traditional internal bond angle coordinates. In original FTST, on the other hand, the phase-space integral is expressed in terms of angular momenta and their conjugate angles, the momentum integrals cannot be done analytically, and the angle variables have no intuitively obvious connection to the system's geometry. The dimension of the FTST phase-space integral is more than twice that of the configuration integral in our new version of CFTST; specifically, the former has dimension $2(N_{q_{int}} + 1)$ and the latter has dimension $N_{q_{int}}$, where $N_{q_{int}}$ is the number of transitional modes. It is thus generally necessary to evaluate the phase-space integral in original FTST with a Monte Carlo method. Since the maximum value of $N_{q_{int}}$ is 5 (for two non-linear top fragments), the configuration integral in our new CFTST can be economically evaluated using an appropriate quadrature scheme. This has the further advantage that V_{tr} needs to be known only on a sufficient number of grid points to converge the integral. The randomly selected geometries arising in a Monte Carlo evaluation necessitate knowing V_{tr} globally and this may involve fitting to a functional form, a task which can be quite difficult in itself.

Our program for improving CFTST has its origins in Ref. 16, where a method for handling the kinetic energy of the transitional/external rotational modes was proposed. This method is rigorous [all internal-external (Coriolis) couplings included] yet leads to simplification (the integrals over the momenta associated with this kinetic energy can be done analytically). The treatment in Ref. 16 was limited to a particular pair of fragments; at atom plus a asymmetric top (a disc). In Ref. 17 (part I), this limitation was removed by treating two dissimilar symmetric tops explicitly and showing how to infer the results for all fragment combinations (any two of atom, linear, top, spherical top, symmetric top, asymmetric top). Since the transitional/external rotational kinetic energy depends only on the types of rigid tops and on their moments of inertia, there is a finite set of phase-space integrals, each of whose momentum integrations can be done analytically. Since all systems fall into one of the fragment combination categories, the new CFTST can be applied to any system without additional formal development. A central feature of Ref. 17 is the restriction to a reaction coordinate connecting the centres of mass of the two fragments. This has the advantage of leading to particularly appealing forms for the transitional/external rotational partition function: it is a compact expression that remarkably factorizes into a purely kinematic factor and a purely configurational factor determined by the potential. The disadvantage is that a centres-of-mass reaction coordinate can be a poor choice in the sense that other choices may lead to a smaller reactive flux.

To address this shortcoming, we have extended, in rigorous fashion, the theory of part I to include all possible reaction coordinates for all fragment combinations. The formal theoretical development accomplishing this task is given in Section 3. The modification to the matrix A_0 of the coefficients of the momenta associated with the transitional/external-rotational kinetic energy for a centres-of-mass reaction coordinate has a simple form [eqn. (37)]. Only the internal kinetic energy quadrant of this matrix is affected, owing to a judicious choice of coordinate systems in Section 3. The modification involves the gradient of F , where F specifies how R_{cm} depends on the internal angles and the value of the chosen reaction coordinate R . The general results of Section 3 are applied only to the simplest possible system atom-diatom, to illustrate the nature of the resulting formula for the transitional/external rotational partition function [eqn. (50)]. This formula is more complicated than its centres-of-mass reaction coordinate counterpart [eqn. (47)] and the complete factorization of kinematic and potential contributions achieved for the latter is not achieved for the former. Nevertheless, the com-

puter time and programming required to evaluate eqn. (50) will be only marginally greater than that to evaluate eqn. (47), so that the computational advantage of our new version of CFTST survives the generalization to arbitrary reaction coordinates. Of course, the computer time required to optimize the reaction coordinate will scale linearly with the volume of the parameter space defining the generalized coordinate. Case-specific results for all fragment combinations and applications to selected systems will be the subject of future publications. It is worth noting that microcanonical sums of states $N(E)$ for optimized reaction coordinates could be readily obtained by inverse Laplace transformation of the transition-state partition function, Q_{TS} , discussed in this paper.

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